Supplementary Material for *Lateral Phishing as a Violation of the Organizational Communication Graph: Cascade Dynamics and Leak-Aware Evaluation*

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This supplement contains secondary tables and figures referenced from the main text. All values are produced by the reproducibility package (`code/results/*.csv`); none are new experiments.

S1. Detector hyperparameters

Model / component	Hyperparameters (fixed, without tuning)
GNN (static / TGN-lite / TGAT-lite)	DIM=24; TGAT: t_{dim}=8; 100 epochs; Adam lr=0.01, wd=$5{\cdot}10^{-4}$; dropout 0.2
GNN classifier head	MLP, 48 hidden units
Tabular detectors (LightGBM)	n_estimators=200–250, seed=42, num_leaves=31, lr=0.1
volumetric / path-based (LightGBM)	n_estimators=200, seed=42; context window W=2 buckets

Table S1: Detector hyperparameters: fixed *uniformly* for all models of a given family, framework defaults or lightly chosen, without grid search. Architecture check (5 seeds, AUC / sensitivity@FPR1%, `exp_cascade_arch.csv`): TGN-lite 0.868/0.108, TGAT-lite 0.884/0.134, hand-crafted context 0.805/0.023—the attention variant of the temporal model matches or exceeds it at every seed.

S2. Cascade detectors and the cascade-model sweep

p_{inf}	0.05	0.1	0.2	0.3	0.5	0.7	0.9
0.05	0.60 [0.57;0.62]	0.74 [0.73;0.76]	0.87 [0.85;0.89]	0.97 [**0.96;0.97**]			
0.1	0.56 [0.53;0.59]	0.77 [0.75;0.78]	0.88 [0.87;0.90]	0.94 [**0.93;0.95**]			
0.2	0.59 [0.57;0.61]	0.77 [0.76;0.79]	0.86 [0.85;0.88]	0.90 [**0.89;0.91**]			
0.3	0.60 [0.58;0.61]	0.78 [0.77;0.80]	0.83 [0.82;0.84]	0.87 [**0.85;0.89**]			
0.5	0.68 [0.67;0.70]	0.78 [0.77;0.79]	0.83 [0.82;0.85]	0.84 [**0.82;0.86**]			
0.7	0.73 [0.73;0.74]	0.80 [0.79;0.81]	0.82 [0.82;0.83]	0.87 [**0.85;0.89**]			
0.9	0.79 [0.79;0.80]	0.84 [0.83;0.85]	0.86 [0.85;0.88]	0.90 [**0.89;0.92**]			

Table S2: Cascade model sweep (AUC, depth 4, 8 seeds; columns 1-hop / manual / volume / temporal): the hierarchy holds across the whole range of the infection probability, the volumetric baseline (with its event-causal counters) is a solid mid-table detector of the *naive* cascade, and the receiving-side signal does not vanish at a realistically low p_{inf}.

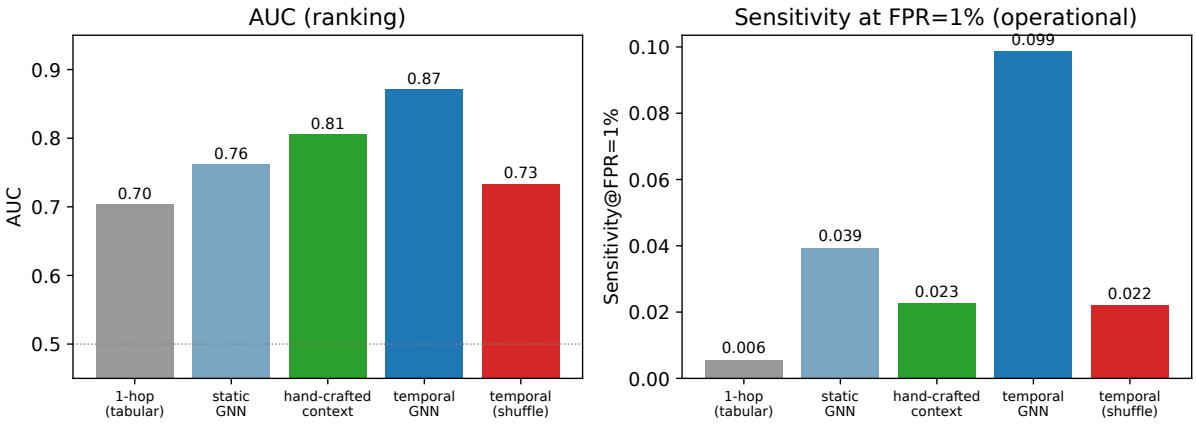


Figure S1: Detector hierarchy on AUC (left) and sensitivity@FPR1% (right); values as in the cascade-detector table of the main text. The temporal GNN dominates; time shuffling (red) levels it with the static GNN.

S3. Inductive validation under increasing scale

	AUC			sensitivity@FPR1%	
Organizations	temp.	manual	volume	temp.	manual
19	0.72	0.79	**0.83**	0.086	0.047
30	0.70	0.77	**0.84**	0.049	0.031
200	0.73	0.82	**0.85**	0.055	0.037

Table S3: Inductive validation on unseen organizations (19 organizations: 5 seeds; 30 and 200 organizations: 3 seeds), with the volumetric baseline in the same pass: volume is the best *transfer* detector (0.83–0.85 AUC, sensitivity 0.24–0.33 at FPR 1%, omitted from the sensitivity columns for width), the temporal detector keeps a 1.8–2.3× operational edge over the hand-crafted context, but neither receiving-side detector transfers across organizations as well as the volumetric signal does.

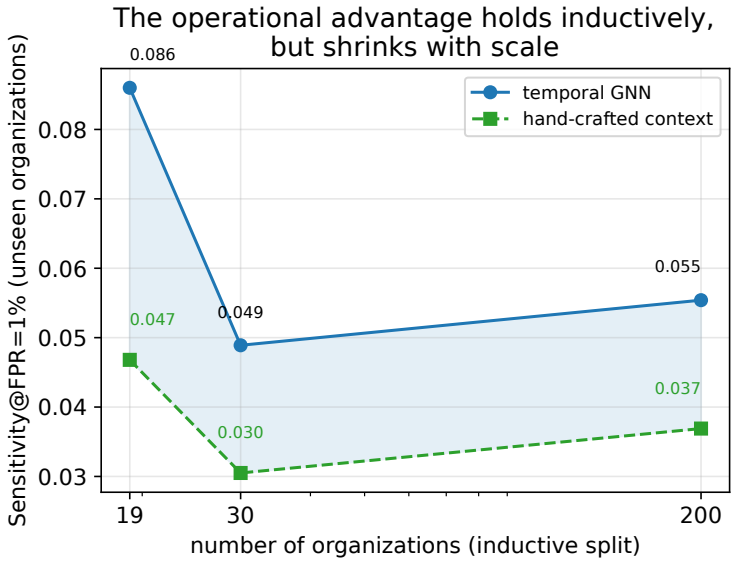


Figure S2: Inductive validation: sensitivity@FPR1% on unseen organizations against the number of companies (values as in Tab. S3). The temporal detector holds a 1.8–2.3× operational edge over the hand-crafted context at every scale; the volumetric baseline (not plotted) transfers best of all.

S4. Replication on real topologies

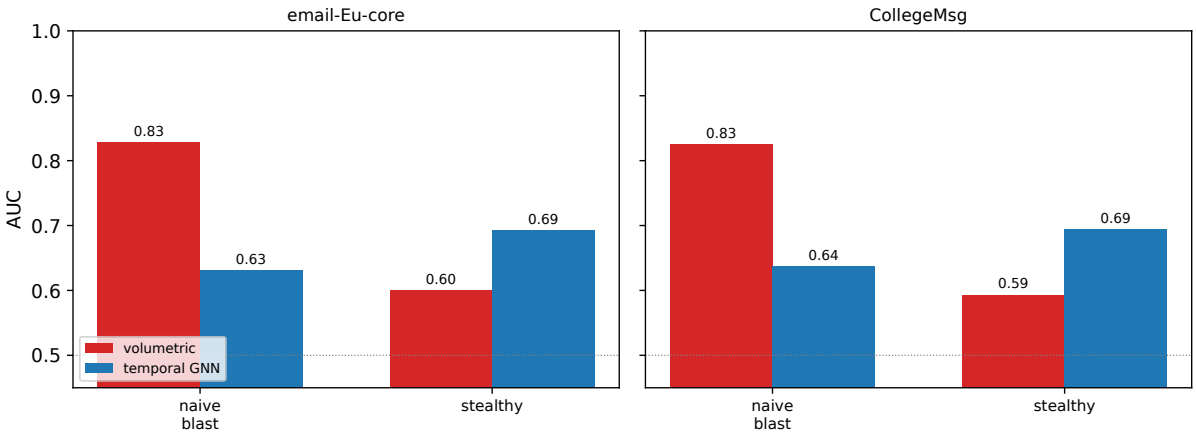


Figure S3: Two real topologies (`email-Eu-core`, `CollegeMsg`). The volumetric detector wins the naive blast (its event-causal counters see the burst) and collapses under the stealthy attack—shown with attack gaps drawn from the background, as in the main table—where the temporal GNN (blue) stays ahead. Values as in block (b) of the real-graph table of the main text.

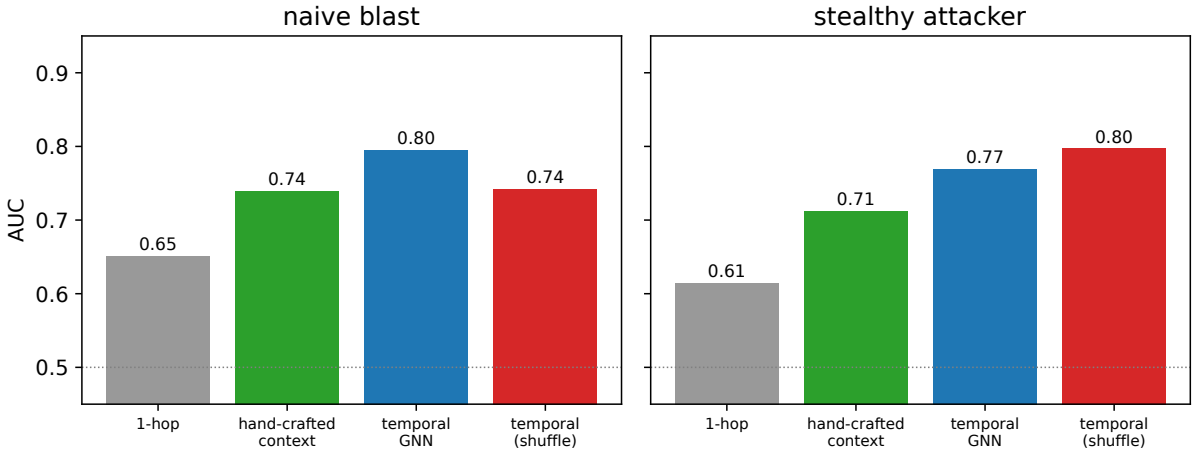


Figure S4: The cascade invariant on Enron, in both regimes, with the volumetric detector included (values as in block (a) of the real-graph table of the main text): under the naive blast the volumetric detector wins (0.87) with the receiving-side hierarchy behind it; under the stealthy attacker, with attack gaps drawn from the background, the receiving-side signal wins (0.77 temporal GNN) while volume stays at 0.67 and the hand-crafted context falls back to the volumetric level (0.71).

S5. Rigor: label noise and the sensitivity–FPR curve

Label noise	vol.	path	path (no precursor)	manual	temp.
0%	0.60 [0.59;0.61]	0.66 [0.65;0.67]	0.63 [0.62;0.64]	0.66 [0.65;0.68]	0.72 **[0.70;0.74]**
5%	0.59 [0.58;0.60]	0.66 [0.65;0.67]	0.63 [0.62;0.64]	0.65 [0.64;0.66]	0.68 **[0.66;0.70]**
10%	0.58 [0.57;0.60]	0.65 **[0.65;0.66]**	0.62 [0.61;0.63]	0.64 [0.63;0.65]	0.64 [0.62;0.66]
20%	0.57 [0.56;0.59]	0.63 **[0.63;0.64]**	0.61 [0.60;0.62]	0.61 [0.59;0.62]	0.59 [0.57;0.60]

Table S4: Audit of noisy training labels (AUC; stealthy attack at K=3 in the second harness (`cascadebench`, 600-account organization; the baseline table of the main text); 20 seeds): the learned temporal model degrades fastest ($0.72 \rightarrow 0.59$), so its advantage shrinks under realistic label noise. Values at K=3 differ from the corresponding row of the baseline table by ≤ 0.02 AUC owing to run-to-run nondeterminism of the temporal model's training (thread scheduling in the GNN step); the qualitative ordering of detectors is unchanged.

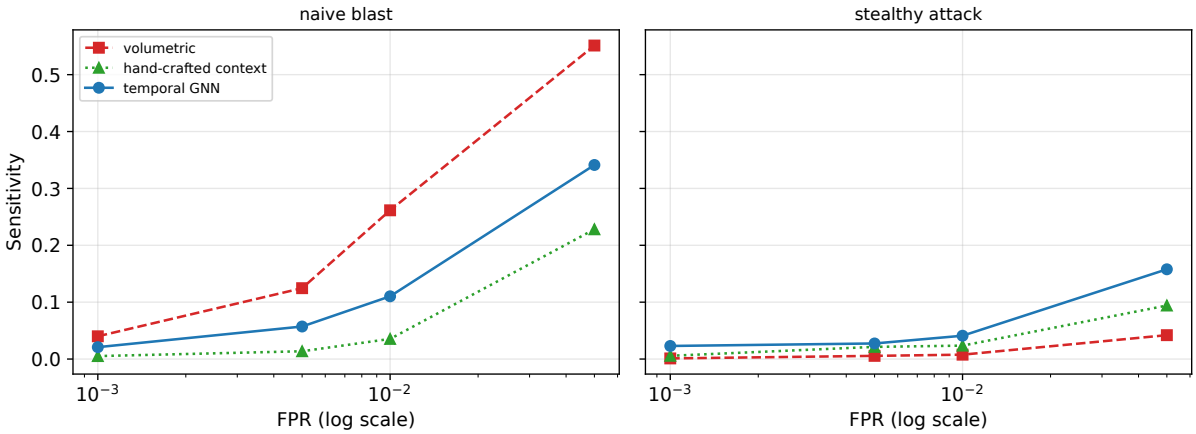


Figure S5: Sensitivity–FPR curve (10 seeds). At low FPR all methods are weak; the volumetric detector leads under the naive blast (its event-causal counters see the burst build up), while the temporal detector stays ahead of it under the stealthy attack.

S6. Feature specification of every detector

All tabular detectors use the same learner (LightGBM, 200–250 trees as in Tab. S1, seed 42); they differ only in their features. Time is divided into buckets b; the context window is $W{=}2$ buckets. $\mathcal{N}(v)$ is the set of contacts of v in the communication graph; $\mathrm{recv}(b)$ and $\mathrm{sent}(b)$ are the sets of accounts that received, respectively sent, mail in bucket b; $\mathrm{rhythm}(s, b)$ is the temporal-consistency bit *tc*. The two experiment harnesses of the package (`experiments/` for the cascade, stealth, scale, and real-graph tables; `cascadebench/` for the baseline, label-noise, and headline-contrast tables) implement identical feature definitions.

- **1-hop (tabular).** For event $s \to v$ in bucket b: $\mathbf{1}[s \in \mathcal{N}(v)]$, $\mathrm{rhythm}(s, b)$, $|\mathcal{N}(v)|$, $|\mathcal{N}(s)|$.
- **Hand-crafted cascade context.** The 1-hop features plus three receiving-side statistics, all computed *event-causally*: an event e at time $t(e) = (b, \text{position in stream})$ sees only events with a strictly smaller $t(\cdot)$—the complete buckets $[b - W, b - 1]$ of the window *and* the earlier events of its own bucket (ties broken by the fixed, shuffled stream order): *burst* $= |(\mathcal{N}(v) \setminus \{s\}) \cap \mathrm{recv}(< t(e))|/|\mathcal{N}(v)|$ (how many of the victim's other contacts were hit strictly earlier within the window); *freshness* $= \mathbf{1}[s \in \mathrm{recv}(< t(e))]$ (the sender was itself hit just before); *propagation* $= |(\mathcal{N}(v) \setminus \{s\}) \cap \mathrm{sent}(< t(e))|/|\mathcal{N}(v)|$ (how many of the victim's other contacts are themselves sending). An anti-leak unit test asserts that every feature of e is unchanged after deleting all events with $t \geq t(e)$ (`tests/test_causality.py`).
- **Volumetric baseline (inspired by COMPA).** Three per-event deviations of the sender from its own profile, computed *event-causally*: recipient novelty $1/(1 + \#\{s \to v \text{ earlier}\})$, time novelty $1/(1 + \#\{\text{events of } s \text{ earlier}\})$, and sender activity $\#\{\text{events of } s \text{ earlier}\}$, where "earlier" covers all earlier buckets *and* the earlier events of the current bucket (in a fixed stream order), so a burst building up within a bucket is visible to the detector. A naive implementation that counts these statistics over the full stream would leak future information into every feature; the counters here are causal by construction. COMPA itself detects compromised accounts on social networks by comparing each new message with a behavioral profile of the account (time of day, source, language, links, topics); we transfer only its per-sender-deviation idea to email headers, restricted to volume and timing.
- **Path-based baseline (inspired by Hopper).** All components are computed event-causally, from events with strictly smaller $t(\cdot)$ within the window $[b - W, b]$: *precursor* $= \mathbf{1}[s \in \mathrm{recv}(< t(e))]$ (the sender was reached just before, i.e. is part of a chain); *momentum* = number of distinct targets of s among strictly earlier events within the window; *rare target* $= 1/(1 + \#\{s' \to v \text{ with } t < t(e)\})$; *unexplained* $= 1 - \mathrm{rhythm}(s, b)$; and the conjunction precursor $\times$ unexplained. Defining momentum over $[b, b{+}W]$ (future buckets), or counting over the full stream, would be a look-ahead leak; the components here are computed directly under the fully event-causal rule with the tie rule above, unifying all detectors. Hopper itself detects lateral movement in enterprise networks from authentication logs, by reconstructing login paths between machines and flagging paths with no benign explanation; it does not process email. We transfer its causal-path idea (a precursor that explains the hop, and a rare target) to the email communication graph.
- **Static GNN.** Node features $X = \log(1{+}[\,|\mathcal{N}(u)|,\ \text{in-degree}(u),\ \text{out-degree}(u)\,])$ (three dimensions, the stream degrees computed from the *training* split only), projected to DIM=24 and propagated through two layers with the normalized adjacency (Eq. (1) of the main text). The head g is an MLP with 48 hidden units and dropout 0.2 over $[Z_s, Z_v, Z_s \odot Z_v, x_{sv}]$, where x_{sv} is the 1-hop feature vector of the event.
- **Temporal GNN (TGN-lite).** Same node features and x_{sv}; a memory of dimension 24 per node, updated *per mini-batch of events in total time order*: the stream is split, within

each bucket, into consecutive chunks of up to 64 events in the fixed (shuffled) stream order; each chunk is *simultaneous*—its events are scored from the memory state before it and cannot see one another—and then the memory of the chunk's recipients is updated with a GRU cell from the mean of the messages $\sigma(W_m m_u)$ of the chunk's senders (Eq. (3)). An anti-leak unit test asserts that the logits of every chunk are unchanged after deleting all later chunks (`tests/test_causality.py`). TGAT-lite replaces the mean pooling by attention with an 8-dimensional time encoding, in the spirit of TGAT (Xu et al., 2020). Training: Adam (lr=0.01, weight decay $5 \cdot 10^{-4}$), 100–120 epochs, binary cross-entropy.

S7. Seed counts per experiment

Experiment	Table / figure	Seeds
Detector hierarchy (case study)	main-text detector table, Fig. S1	5
Library TGN (extra row)	main-text detector table	3
Architecture robustness (TGN-lite / TGAT-lite)	Sec. S1	5
Stealthy attacker sweep	main-text stealth table and figure	20
Baselines (a) stealth / (b) surgical	main-text baseline table	20
Legal broadcast waves (easy + hard, default + stealth)	main-text baselines section	20
Headline contrasts (Wilcoxon, Cliff's δ; both benign protocols)	main-text rigor section	20
Injection-signature control (KS, probe, empirical)	main-text rigor table	3
Label-noise audit	Tab. S4	20
Static-blindness audit (compromised senders)	—	20
Cascade-model sweep	Tab. S2	8
Inductive validation (19 / 30 / 200 org.)	Tab. S3, Fig. S2	5 / 3 / 3
Sensitivity–FPR curve	Fig. S5	10
Enron cascade	main-text real-graph table (a), Fig. S4	8
SNAP graphs (`email-Eu-core`, `CollegeMsg`)	main-text real-graph table (b)	3

Table S5: Number of random seeds behind every reported aggregate. The stealthy, baseline, label-noise, and contrast experiments use 20 seeds; the detector hierarchy and architecture checks 5; real-graph replications use 8 (Enron) and 3 (SNAP), and are reported as directional confirmations.